PROJECT PROPOSAL

EV Battery Health Prediction Using Limited and Noisy Sensor Data

Special Project • Machine-Learning-Based EV Battery Health & Lifespan Predictor

. Project Title

. Introduction

EV batteries lose capacity as they age. State of Health (SOH) is current capacity ÷ initial capacity × 100. Remaining Useful Life (RUL) is the number of charge–discharge cycles left until SOH reaches 80%, the retirement point used in your draft. These definitions vary between papers, so this project fixes them upfront.

Machine learning can learn degradation patterns from voltage, current and temperature data. But most models are developed on lab test data. In real vehicles, field data is affected by noise, missing values and low sampling frequency. This project studies how well ML health predictors hold up as data quality worsens, using public lab datasets and a data-quality simulator.

. Problem Statement

Given only limited and noisy sensor data from a battery, how accurately can ML estimate SOH and RUL for a battery it has never seen? At what data quality does the prediction stop being reliable?

. Research Gap

In simple terms:

Models are trained and tested on near-perfect data, but real vehicles cannot provide it.

The closest study simulated poor data at a single fixed level, on private cells, so nobody else can reproduce or extend it.

Benchmarks like BatteryML and BatteryLife standardise data and models. In what I reviewed, they do not systematically test data degradation.

Many studies still test on cycles from batteries already seen in training, which can overstate performance on a new battery.

. Objectives

Build a clean, reproducible SOH and RUL baseline on public data, tested on unseen batteries.

Build a data-quality simulator with four controlled degradation factors.

Measure how prediction error grows as data worsens, and find the minimum data quality that still meets a chosen error tolerance.

Find the point where the model stops beating a “cycle count only” baseline.

Test which feature groups and models are most robust, and whether training on degraded data helps.

Package the findings in a simple demo app and report.

. Proposed Solution

A data-quality stress-test pipeline:

Train baseline models on clean data.

Worsen the test data step by step with the simulator.

Record error at each level.

Report minimum data requirements, break-even points and robust features.

Show it all in a small interactive demo.

. Proposed Methodology

Pipeline: Raw lab data → cycle segmentation → data-quality simulator → feature extraction → models → cross-battery evaluation → outputs and demo.

Modules: Data Loader, Data-Quality Simulator, Feature Extractor, Model Trainer, Evaluator, Demo App.

Data-quality factors (levels to be tuned after exploring each dataset):

For realistic anchors, Wang et al. used 1 Hz lab data reduced to one sample per 10 s, with voltage precision of about 1 mV and current of about 0.1 A.

Steps:

Weeks 1–2: Literature reading, download data, exploratory analysis (native sampling rate, cycle counts, capacity curves).

Weeks 3–4: Cycle segmentation, SOH/RUL labels, feature extraction, clean baseline.

Weeks 5–6: Build the simulator and run all degradation sweeps.

Weeks 7–8: Compare models and features, and test training on degraded data.

Week 9: Streamlit demo.

Week 10: Report and viva preparation.

Evaluation:

Leave-one-battery-out testing.

MAE and RMSE in SOH percentage points, and MAE in cycles for RUL.

Comparison against the cycle-count-only baseline.

A tolerance threshold (for example, a few SOH points), chosen and justified by you before running experiments.

Features (per cycle, from the charge segment where possible):

Time to move between voltage levels

Voltage statistics

Temperature maximum and rise

Charge-based capacity estimate

Optional incremental-capacity peak features

. Dataset

NASA PCoE (main). Four widely used 18650 cells (B0005, B0006, B0007, B0018). They were charged at 1.5 A to 4.2 V, then held at constant voltage until the current dropped to 20 mA, and discharged at 2 A. Cycle counts are 168 for three cells and 132 for B0018. The raw files include time, voltage, current and temperature vectors, with capacity for discharge cycles. A second group tested at 43°C can add more cells.

CALCE CS2 (second dataset). Reported as 1.1 Ah graphite/LiCoO■ cells, charged at 0.5C to 4.2 V then held at constant voltage, with 1C discharge.

Oxford (optional). Pouch cells tested at 40°C in a thermal chamber with a drive-cycle derived from the urban Artemis profile, measured every 100 cycles. That gives few capacity labels per cell, so it suits at most a side experiment.

MIT/Stanford (optional, large). The files are 2–3 GB each. Use it only if time permits, perhaps via BatteryML.

Two cautions:

Sources disagree on cell chemistry (NASA is described as LiCoO■ in one paper and NCA in another, and CALCE similarly). Verify with the official dataset documentation.

The independent unit is the battery, not the cycle. The 2026 study stresses that four NASA cells means four independent units, not 636 cycles. Report results per battery and treat them as descriptive.

. Machine Learning Techniques

Baselines: Cycle-count-only model and Ridge Regression.

Main models: Random Forest and XGBoost (or scikit-learn Gradient Boosting) for SOH, and the same tree models for direct RUL regression.

Validation: Leave-one-battery-out cross-validation.

Robustness: Noise-based data augmentation (training on degraded data).

Interpretation: Permutation importance, with SHAP optional.

Optional only: A small LSTM if time remains. It is not needed for the core result.

. Software / Tools

Language and libraries: Python 3.10+, NumPy, pandas, SciPy (to read .mat files), scikit-learn, XGBoost, Matplotlib or Plotly.

Environment: Jupyter or Google Colab, Git.

Demo: Streamlit.

Optional: SHAP and BatteryML.

Hardware: An ordinary laptop. No EV, sensors or GPU needed.

. Expected Output

Cleaned, documented datasets and SOH/RUL labels.

Trained models and a leave-one-battery-out results table.

Robustness curves (error versus data quality, per factor).

A “minimum data quality” summary and a break-even analysis versus the cycle-count baseline.

A Streamlit demo with data-quality sliders that shows predicted SOH, RUL and health grade. The grade thresholds are your own design choice, such as ≥90% Good, 80–90% Moderate and <80% Replace soon.

Report and viva slides.

. Expected Results

No accuracy is claimed before training. These are hypotheses to test, and a negative result is still valid:

Error will grow as data worsens, but not equally for all four factors.

Some feature groups will tolerate degradation better than others.

Training on degraded data may recover part of the lost accuracy.

A break-even point versus the cycle-count baseline may appear.

The clean baseline will likely show larger errors on unseen batteries than on same-battery splits, consistent with prior work.

. Limitations

All data is from lab cells, not real vehicles. Simulated degradation only approximates field conditions.

The number of independent batteries is small, so results are descriptive.

Cell-level data only, with no pack-level behaviour.

Results may not transfer to other chemistries.

SOH/RUL definitions and the tolerance threshold are project choices.

This is a research study, not a safety-certified tool.

. Future Scope

Validate on open real-vehicle data, such as the field dataset disclosed with Liu et al. (2025), if accessible.

Add uncertainty ranges (Gap 4).

Scale up with BatteryML or BatteryLife datasets.

Explore transfer across chemistries.

Only later consider embedded deployment.

. References

Verified through the sources reviewed. Add DOIs and adjust the format to your institute's style.

. A. Severson et al., "Data-driven prediction of battery cycle life before capacity degradation," Nature Energy, 4(5), 383–391, 2019. doi:10.1038/s41560-019-0356-8

. Zhang, X. Gui, S. Zheng, Z. Lu, Y. Li, J. Bian, "BatteryML: An Open-source Platform for Machine Learning on Battery Degradation," ICLR 2024.

R. Tan et al., "BatteryLife: A Comprehensive Dataset and Benchmark for Battery Life Prediction," KDD 2025. doi:10.1145/3711896.3737372 . Wang, C. Zhang, X. Meng, L. Zhang, X. Li, W. Zhang, "A Novel Feature Engineering-Based SOH Estimation Method for Lithium-Ion Battery with Downgraded Laboratory Data," Batteries, 10(4), 139, 2024. doi:10.3390/batteries10040139

. Barua, M. D. Collins, M. S. Islam, A. Barua, K. N. A. Siddiquee, "Predictor Structure Modulates Validation Inflation and Cell-Transfer Reliability in Battery State-of-Health Estimation," Batteries, 12(9), 342, 2026. doi:10.3390/batteries12090342

. Liu et al., "Multi-modal framework for battery state-of-health evaluation using open-source electric vehicle data," Nature Communications, 16, 1137, 2025.

. Lanubile et al., "Domain knowledge-guided machine learning framework for state of health estimation in lithium-ion batteries," Communications Engineering, 3, 168, 2024.

. Roman, S. Saxena, V. Robu, M. Pecht, D. Flynn, "Machine learning pipeline for battery state-of-health estimation," Nature Machine Intelligence, 3, 447–456, 2021.

. Paneru et al., "Remaining Useful Life Prediction for Batteries Utilizing an Explainable AI Approach with a Predictive Application for Decision-Making," arXiv:2409.17931, 2024.

. Saha, K. Goebel, "Battery Data Set," NASA Ames Prognostics Data Repository, 2007.

. Howey, C. Birkl, "Oxford Battery Degradation Dataset 1," University of Oxford, 2017. doi:10.5287/bodleian:KO2kdmYGg

CALCE Battery Research Group, University of Maryland, public battery data (CS2). Original dataset papers are He et al. (2011) and Xing et al. (2013), as cited in [2]. Copy the full titles from [2]'s reference list.

Data-Quality Factors

Factor	What it Mimics	Example Levels
Sampling interval	Slower BMS logging	1st, 2nd, 5th, 10th, 30th reading
Noise and rounding	Cheaper/coarser sensors	Add noise to V/I/T and round readings
Missing data	Dropped messages	0/5/20/40% removed + gaps
Incomplete charge	Partial charging	Full curve / upper half / narrow window